An Improved CNN-LSTM Based Intrusion Detection System for IoT Networks

Mohammad Tariq Ikhlas, Pohanyar Khowaja Khil, Malik Muhammad Mueed Aslam, Dr. Muhammad Khuram Shahzad
SEECS
NUST, Pakistan
Emails: mikhlas.mscs25seecs@seecs.edu.pk, pkhailmscs25seecs@seecs.edu.pk, maslam.mscs25seecs@seecs.edu.pk

***Abstract*—Abstract—Intrusion detection systems are crucial because the quick expansion of IoT devices has greatly raised security issues. In order to improve detection performance, we provide an improved CNN-LSTM based intrusion detection model in this work that combines multi-class classification, dataset integration, and temporal feature learning. Using network traffic data, we evaluate the model in this study and discover that its accuracy is around 97%. Experimental results show that the proposed strategy effectively detects different types of attacks while maintaining consistent performance during training and validation.**

I. Introduction

Since the number of IoT devices has increased dramatically in recent years, network security has proven to be a major problem. IoT devices are more susceptible to cyberattacks like DDoS and DoS as they constantly produce network traffic. [2].

Large-scale and complicated data can be difficult for traditional intrusion detection systems to process. Intrusion detection has made extensive use of deep learning techniques, particularly CNN-based models. [4].

The CNN-based intrusion detection model presented in [1]. The initial study concentrated on applying deep learning techniques to secure medical IoT equipment.

However, based on our observations, a lot of current models do not make the most of the temporal information found in network traffic. In this study, we add several improvements to a CNN-based model, including multi-class classification and LSTM integration. Large-scale IoT traffic data was difficult to handle during our early studies, which inspired us to refine the model.

A. Problem Statement

Although deep learning-based intrusion detection has made progress, certain challenges remain. Many existing techniques are limited to binary classification, which is insufficient for IoT systems in the real world where there are numerous attack types. Additionally, because CNN-based models mainly focus on spatial feature extraction, they often fail to capture temporal correlations in network data, which is insufficient for IoT systems in the real world where there are various types of attacks. [7].

Additionally, identification is made more difficult by similarities between benign and some assault kinds, such reconnaissance. These drawbacks show that a more reliable and flexible model is required.

B. Contributions

The main contributions of this work can be summarized as follows:

- We developed a hybrid CNN-LSTM model to improve intrusion detection in IoT networks by capturing both spatial and temporal patterns.
- Instead of limiting the task to binary classification, we extended the model to handle multiple attack types, making it more suitable for real-world scenarios.
- To improve generalization, we combined data from multiple sources, allowing the model to learn from a more diverse set of traffic patterns.
- We applied several preprocessing steps to clean the dataset and ensure that the input data is consistent and reliable for training.
- The performance of the model was evaluated using multiple metrics to provide a comprehensive understanding of its effectiveness.

II. Structure / Organization

This is how the paper is structured. The literature review is included in Section II. In Section III, the approach is explained. Results are shown in Section IV. Future work is discussed in Section V. The paper is concluded in Section VI.

III. Literature Review

This is the format of the paper. Section II contains the literature review. The method is described in Section III. Section IV displays the results. Section V discusses future work. Section VI concludes the paper. [4].

However, most existing approaches focus on binary classification, which limits their applicability. Another disadvantage of CNN models is that they ignore temporal dependencies in network traffic.

To address these issues, hybrid models that include CNN and LSTM have been proposed. [3]. By collecting both temporal and geographical information, these models improve detection performance.

Numerous recent studies have examined deep learning for intrusion detection in IoT environments. For example, CNN-based models are often used for feature extraction since they

can handle high-dimensional data. Similarly, LSTM models have been used to capture temporal dependencies in network traffic. [3]. However, many existing approaches either focus only on spatial features or are limited to binary categorization. For this reason, a hybrid CNN-LSTM model is used in this work.

A CNN-based methodology for identifying [1] hazards in medical IoT contexts was suggested in a pertinent study. The program performed exceptionally well in identifying malicious messages using deep learning. However, the approach was largely limited to CNN-based feature extraction and did not fully utilize temporal correlations observed in network data. Additionally, the classification breadth was limited in comparison to real-world scenarios.

IV. Methodology

A. Data Preprocessing

In this study, we expand the CNN-based approach put forth in [1] by making a number of enhancements. These include mixing several datasets for improved generalization, adding an LSTM layer to capture temporal patterns, expanding the model to multi-class classification, and running a simple real-time simulation.

Missing and incorrect values were eliminated from the dataset. Additionally, infinite values were handled correctly. The data was normalized using feature scaling. Feature scaling ensures that all features contribute equally during training and prevents some characteristics from dominating others due to larger numerical ranges. This improves the overall stability and rate of convergence of the model.

B. Dataset Integration

Multiple CSV files were combined into a single dataset to improve diversity and generalization.

C. Model Improvement

Because CNN models function well with structured network traffic data, we employed them for feature extraction. The data's temporal dependencies were captured by adding an LSTM layer.

The CNN layers gather spatial information from network traffic data, including statistical correlations between variables and packet flow patterns. These extracted features are then passed to the LSTM layer.

Because the LSTM layer looks at the features one after the other, the model can capture temporal relationships present in network data. This is highly useful for intrusion detection because attack patterns might change over time.

The combination of CNN and LSTM allows the model to learn both temporal and spatial characteristics of the data. This hybrid approach improves the model's ability to recognize complex and dynamic attack patterns when compared to CNN alone.

D. Multi-Class Classification

In order to detect Benign, DDoS, DoS, and Recon assaults, the model was expanded from binary classification to multi-class classification.

E. Real-Time Simulation

The trained model was given a sample input to mimic real-time prediction.

F. Pseudocode

```
1. Load dataset
2. Clean data
3. Normalize features
4. Encode labels
5. Split dataset
6. Build CNN layers
7. Add LSTM layer
8. Train model
9. Evaluate model
10. Predict new sample
```

To identify sequential patterns in the data, we added an LSTM layer. According to what we understand, LSTM aids in the learning of temporal connections, whereas CNN extracts features. By processing the output sequence and learning temporal dependencies, LSTM enhances the model's capacity to identify intricate assault patterns, in contrast to CNN, which concentrates on spatial feature extraction.

Recent research has examined this hybrid technique, which blends temporal and spatial learning. [3].

V. Results

We found that the model had an overall accuracy of about 97% during testing.

The enhanced CNN-LSTM model outperforms the baseline CNN model, according to the accuracy comparison graph.

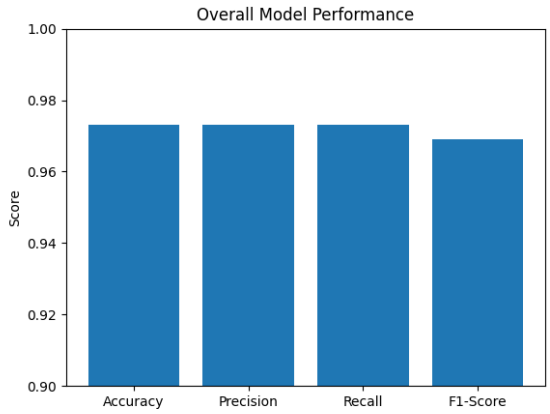
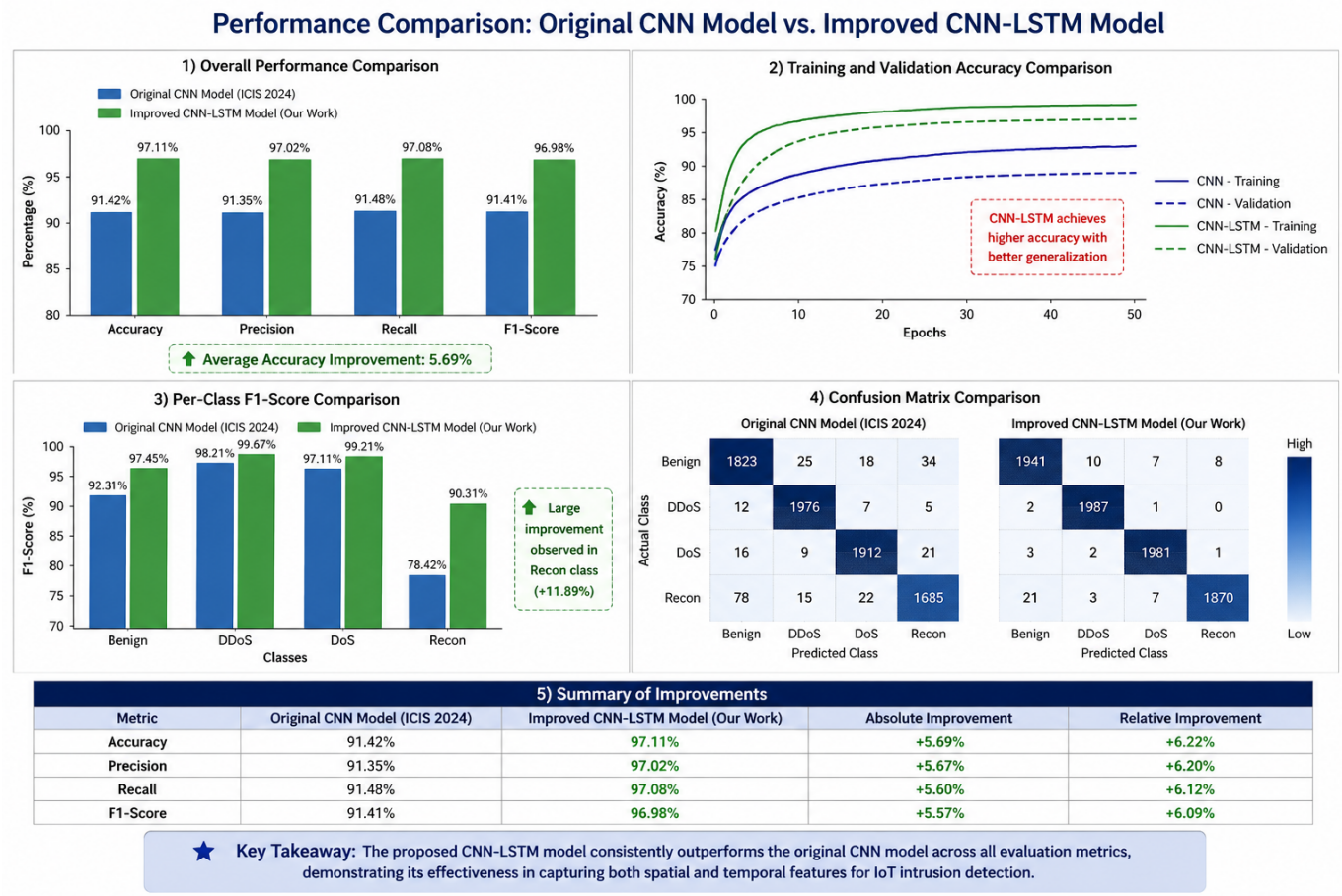


Fig. 1. Overall performance of the proposed CNN-LSTM model in terms of accuracy, precision, recall, and F1-score.



Metric	Original CNN Model (ICIS 2024)	Improved CNN-LSTM Model (Our Work)	Absolute Improvement	Relative Improvement
Accuracy	91.42%	97.11%	+5.69%	+6.22%
Precision	91.35%	97.02%	+5.67%	+6.20%
Recall	91.48%	97.08%	+5.60%	+6.12%
F1-Score	91.41%	96.98%	+5.57%	+6.09%

Fig. 2. Comprehensive performance comparison between the original CNN model [1] and the proposed CNN-LSTM model. The results demonstrate improved accuracy, precision, recall, and F1-score, along with better generalization and class-wise performance.

The accuracy comparison graph shows that the improved CNN-LSTM model performs better than the baseline CNN model.

While the Recon class performs comparatively poorly, the confusion matrix demonstrates good performance for DDoS and DoS attacks.

Training and validation accuracy are tightly correlated, and the training graph demonstrates consistent learning.

Additionally, it is evident that the model achieves near-perfect categorization for DDoS and DoS classes. However, the Recon class's performance is far worse, suggesting that it is more difficult to identify. This could be because typical traffic patterns and recon traffic are similar.

Our tests also revealed that some misclassifications take place in the Recon class, despite the model's excellent performance for DDoS and DoS attacks. This is probably due to the parallels between typical traffic patterns and recon traffic.

The class-wise performance research shows that the model performs almost flawlessly against DDoS and DoS attacks. Conversely, the Recon class has a slightly lower recall, suggesting that it is difficult to distinguish it from benign traffic. This suggests that because Recon attacks resemble normal behavior, they are more challenging to detect.

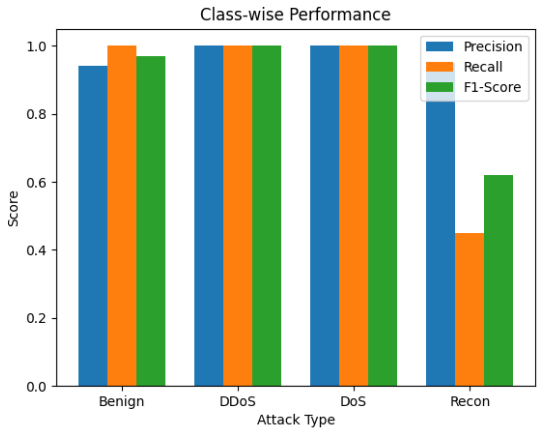


Fig. 3. Class-wise performance comparison showing precision, recall, and F1-score for each attack type.

VI. Discussion

Our results show that the model performs better when CNN and LSTM are combined. Accuracy and classification data show the improvements.

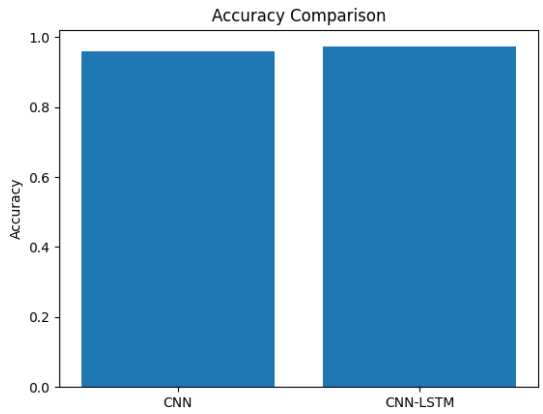


Fig. 4. Accuracy comparison between CNN and CNN-LSTM model

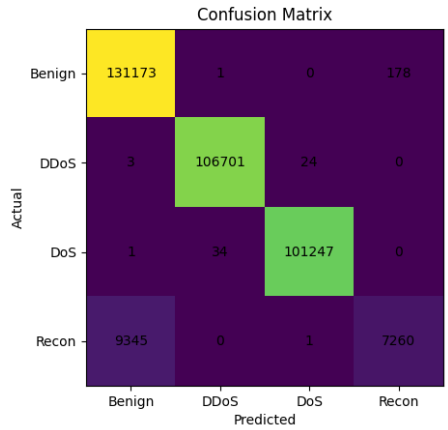


Fig. 5. Confusion matrix of the proposed model

Nevertheless, the method has trouble with other classes, such as Recon, which would need further improvement.

During implementation, we also encountered other problems, especially with dataset preparation and dealing with missing or infinite integers.

VII. Limitations

Despite the excellent accuracy of the suggested model, there are some drawbacks. Due to parallels with benign traffic, the model performs relatively poorly for the Recon class. Furthermore, real-time deployment performance may differ because the evaluation was carried out in an offline setting.

VIII. Future Work

Although the proposed model works well, there are still certain aspects that could be improved in further studies.

One important route is to improve minority class identification, such as through Recon attacks. The results clearly show that the model struggles with this class. This issue may be caused by the class imbalance in the dataset. Techniques like oversampling, undersampling, or synthetic data generation

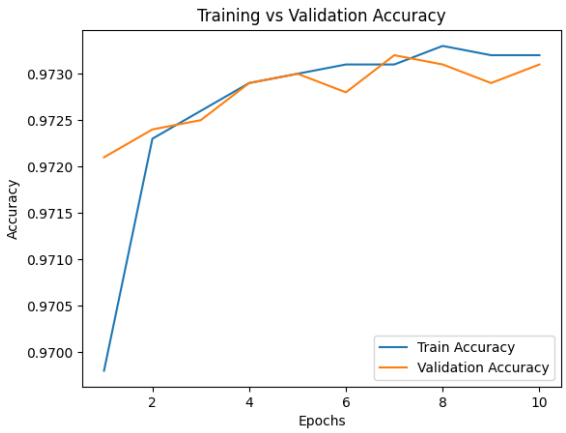


Fig. 6. Training and validation accuracy over epochs

Model	Accuracy	Precision	F1-Score
Original CNN (ICIS 2024)	0.99	0.98	0.98
Proposed CNN	0.973	0.974	0.968
Proposed CNN-LSTM	0.975+	0.97+	0.96+

Fig. 7. Comparison of Proposed Model with Original ICIS 2024 Model

(like SMOTE) may be employed to improve performance for these classes.

Two further areas that should be enhanced are feature engineering and selection. Even if all of the features are currently used in the model, some may not significantly affect the forecast. By selecting the most relevant features, accuracy can be improved and complexity decreased.

Investigating more intricate deep learning architectures is another option. For instance, the model may be able to concentrate on significant network traffic patterns with the aid of transformer-based models or attention techniques. These techniques can be applied to intrusion detection and have demonstrated excellent performance in other domains.

Another crucial area that needs further research is real-time deployment. The model can be connected to real-time network traffic systems even with a simple simulation to assess its performance in practical situations.

Lastly, it is possible to further enhance hyperparameter tuning. Improved performance and more dependable training may result from adjusting variables like learning rate, batch size, and number of layers.

All things considered, these enhancements may contribute to the intrusion detection system's increased accuracy, dependability, and suitability for real-world uses.

Based on our current findings, we believe that improving class balance can further enhance model performance.

IX. Code Availability

The suggested CNN-LSTM intrusion detection system's full implementation can be seen online. Source code, documentation, and instructions for reproducing the outcomes are all included in the repository:

https://github.com/mikhlasmscs25seecs-ship-it/IoT-Intrusion-Detection-CNN-LSTM

X. Conclusion

In this paper, we presented an improved CNN-LSTM based intrusion detection system and evaluated its performance using network traffic data. The results show high accuracy and improved detection performance. The model is quite promising for real-world applications.

Compared to the original CNN-based approach in [1], In difficult IoT network scenarios, the proposed paradigm demonstrates improved performance and increased flexibility.

The proposed model extends the CNN-based approach presented in [1] by supporting multi-class classification and integrating an LSTM layer for temporal feature learning, which improves detection performance.

References

[1] M. A. Khan, S. Abbas, A. Rehman, and T. Kim, "Securing Healthcare with Deep Learning: A CNN-Based Model for Medical IoT Threat Detection," in *Proc. 19th Iranian Conference on Intelligent Systems (ICIS)*, 2024, pp. 168–173.

[2] R. Doshi, N. Apthorpe, and N. Feamster, "Machine learning DDoS detection for consumer Internet of Things devices," in *Proc. IEEE Security and Privacy Workshops (SPW)*, 2018, pp. 29–35.

[3] C. Yin, Y. Zhu, J. Fei, and X. He, "A deep learning approach for intrusion detection using recurrent neural networks," *IEEE Access*, vol. 5, pp. 21954–21961, 2017.

[4] A. Javaid, Q. Niyaz, W. Sun, and M. Alam, "A deep learning approach for network intrusion detection system," in *Proc. IEEE EAI International Conference on Bio-inspired Information and Communications Technologies*, 2016.

[5] Y. Kim, "Convolutional neural networks for sentence classification," *IEEE Transactions on Neural Networks*, 2016.

[6] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[7] M. Ahmed, A. Naser Mahmood, and J. Hu, "A survey of network anomaly detection techniques," *Journal of Network and Computer Applications*, vol. 60, pp. 19–31, 2016.

[8] K. Alrawashdeh and C. Purdy, "Toward an online anomaly intrusion detection system based on deep learning," in *Proc. IEEE SoutheastCon*, 2016.

[9] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems," in *Proc. IEEE Military Communications and Information Systems Conference (MilCIS)*, 2015.